

Product Requirements Document (PRD): Kadhambam

1. Product Overview & Purpose

Kadhambam is a modern, premium e-commerce web application specializing in high-quality dry fruits and nuts. The platform is designed to provide a seamless, visually elegant shopping experience for customers while minimizing backend maintenance and operational costs through a strictly serverless architecture.

The primary goals are:

- To showcase organic, premium products using an earthy, high-end visual aesthetic.
- To facilitate a frictionless, zero-fee payment process using a static UPI QR flow.
- To empower store administrators with a secure, highly automated dashboard for managing inventory, verifying payments, and communicating with customers directly via WhatsApp.

2. Technical Architecture

The application is built on a 100% serverless, Backend-as-a-Service (BaaS) model to ensure high performance, easy scaling, and zero server maintenance.

- **Frontend Framework:** React.js (built with Vite), HTML, CSS, JavaScript (Standard React Hooks).
- **Authentication:** Firebase Authentication (Email/Password & Google Sign-In).
- **Database:** Firebase Firestore (NoSQL).
- **Media Storage:** Cloudinary (Direct client-side uploads via Unsigned Upload Preset API).
- **Hosting & Deployment:** Cloudflare Pages.

3. Customer Storefront Features (User-Facing)

3.1. Navigation & Header

- **Branding:** Left-aligned custom "Kadhambam" logo.
- **Navigation:** Centered links (Home, Shop, Categories, About Us).
- **User Actions:** Right-aligned Login/Profile button and a Cart icon featuring a dynamic item count badge.

3.2. Home Page

- **Hero Carousel:** A constrained-width image slider (not full screen) featuring prominent curved edges (e.g., `border-radius: 24px`) and margins on the left/right. It will cycle through 4 promotional images with visible navigation indicators (dots/arrows).
- **Premium Aesthetic:** Earthy color palettes (warm browns, soft greens, off-whites) to reinforce the organic nature of the products.

3.3. Shop & Product Browsing

- **Filtering:** A sleek filter bar or sidebar allowing users to sort by dynamic categories (e.g., Almonds, Cashews, Mixed Berries).
- **Product Grid:** High-end product cards featuring soft drop shadows, rounded corners, high-quality images, and elegant typography.
- **Interaction:** A stylish "Add to Cart" button with smooth hover animations on each product card.

3.4. About Us Page

- **Brand Storytelling:** A 4-block alternating split-screen layout (Image Right/Text Left, then Image Left/Text Right).
- **Visuals:** Image placeholders feature matching curved corners to maintain design consistency across the site.

3.5. Checkout & Payment Flow

- **Order Summary:** A clear breakdown of cart items and the total price.
- **Static UPI Payment:** A dedicated block displaying a static UPI QR code and a text VPA ID (e.g., `store@upi`).
- **Verification Input:** An input field requiring the user to enter their **UTR Number (Unique Transaction Reference)** post-payment.
- **Success State:** Upon submitting the UTR, the user is redirected to a minimalist success screen displaying exactly: **"We will confirm your payment in a while."**

4. Secure Admin Panel Features (Internal)

4.1. Security & Access

- **Protected Routes:** The `/admin` path is heavily guarded, accessible only to users with verified admin credentials in Firestore.

4.2. Dashboard & Order Management

- **Order Table:** Displays pending and historical orders (Order ID, Date, Customer Details, Items, Total, UTR Number, and Status).
- **Payment Verification:** A "Verify Payment" action button that updates the Firestore order status from "Pending" to "Verified".
- **Automated WhatsApp Integration:** A "Send WhatsApp" button next to each order. It sanitizes the customer's phone number (stripping non-numeric characters) and triggers a [wa.me](#) URL to open a pre-filled message: *"Hello [Customer Name], your payment verification is successful and your order ([Order ID]) is prepared for delivery."*

4.3. Catalog Management

- **Categories:** Full CRUD (Create, Read, Update, Delete) control over product categories.
- **Products:** Add/Edit/Delete functionality for the inventory. Features a drag-and-drop zone that connects directly to the Cloudinary API for seamless image hosting without routing through a backend server.

4.4. Data Management & Export

- **CSV Downloads:** Prominent buttons to instantly parse and download [users](#) and [orders](#) Firestore collections as [.csv](#) files directly to the admin's local machine.
- **Automated Data Retention:** To stay strictly within Firebase free-tier limits, a silent background function runs upon Admin Login. It queries orders and users older than 30 days (capped at 50 documents per batch) and permanently deletes them using a Firestore [writeBatch](#).